

BTECCCE25101: Engineering Calculus and Applications

Course Type	BSC	Semester	I
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Teaching Scheme		Credits		Examination Scheme	
Lecture	03Hr./Week	Lecture	03	CIE Marks	
Tutorial	01Hr./Week	Tutorial	01	ESE Marks	
Practical/Studio	00Hr./Week	Practical/Studio	00	Total Marks	100
Total	04Hr./Week	Total	04		

Course Description

This Applied Mathematics course will explore the practical applications of calculus, focusing on topics like applications of derivatives, successive differentiation, partial derivatives, ordinary differential equations, multiple integrals, and Fourier series. It aims to equip students with the mathematical tools to model and solve problems in various scientific and engineering fields.

Course Outcomes

CO No.	Statement
1	Apply the knowledge of differential calculus to solve problems arising in various fields of engineering and technology
2	Learn the notion of partial differentiation to calculate rates of change of multivariate functions and Applications
3	Solve ordinary differential equations; apply the knowledge of the same related to simple electrical circuits, orthogonal trajectory, and Newton's Law of cooling.
4	Evaluate double & triple integrals. Apply knowledge of multiple integrals to find area and volume.
5	Calculate the Full range and half range Fourier series, and apply Fourier series in harmonic analysis and applications.

Mapping of COs to POs and PSOs

CO Number	POs												PSOs				BTL
	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3	PSO 4	
CO1	2	2	1			1						1	1			1	1,2,3
CO2	2	1							2			1	1			1	2,3
CO3	3	1		1								1	1			1	1,2,3
CO4	3	2	1	2		1			1			1	1			1	1,3
CO5	3	2	1	2		2			1			1	1			1	1,2,3

Affinity Level: 1- Slight, 2- Moderate, 3-Substantial, BTL: Bloom's Taxonomy Level

COURSE CONTENT

UNIT NO. 01	DIFFERENTIAL CALCULUS	08 Hours	CO	BTL
Introduction to limits and continuity, Successive differentiation, Leibnitz's Theorem for n^{th} derivatives of the product of two functions, Rolle's theorem, Lagrange's and Cauchy's mean value theorem, Taylor's and Maclaurin's series expansion.			1	1,2,3
UNIT NO.02	PARTIAL DERIVATIVES	10 Hours		
Multivariable functions, Introduction to Partial derivatives, Differentiation treating one variable as constant, Composite functions, Total Derivatives, Euler's Theorem on Homogeneous Functions, Applications-Jacobians, functionally dependence/independence.			2	2,3
UNIT NO.03	ORDINARY DIFFERENTIAL EQUATIONS	10 Hours		
Definition, Order & Degree of DE., Exact differential equations, Equations reducible to exact equations, Linear differential equations, Reducible to linear differential equations (Bernoulli's equation). Applications of ODE -Orthogonal trajectories, Newton's law of cooling, Law of Decay, Electrical Circuits etc.			3	1,2,3
UNIT NO.04	MULTIPLE INTEGRATION	07 Hours		
Introduction to 2D and 3D coordinate systems, Evaluation of double integration, Evaluation of double integrals by changing the order of integration and by changing into polar coordinates, Applications of Double integration. Introduction to triple integration. Applications			4	1,3
UNIT NO.05	FOURIER SERIES AND HARMONIC ANALYSIS	10 Hours		

FOURIER SERIES: Periodic functions, Dirichlet's condition, Fourier Series of periodic functions with period 2π and with arbitrary period $2L$, Odd and even functions, Half-range Fourier series, Harmonic Analysis, Applications.	5	1,2,3
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Textbooks

1	Higher Engineering Mathematics, B.S. Grewal, Khanna Publishers, 35th Edition, 2000.
2	Higher Engineering Mathematics, Ramana B.V., Tata McGraw Hill New Delhi, 11 th Reprint, 2010.

Reference Books / Journal Articles / Weblink

Reference Books	
1	Applied Mathematics (Vol I & II) by P. N. Wartikar & J. N. Wartikar
2	Advanced Engineering Mathematics, 7 th Edition, By Peter V. O'Neil (Thomson learning publisher)
3	Erwin Kreyszig, Advanced Engineering Mathematics, 9th Edition, John Wiley & Sons, 2006.

Relevant Springboard Courses

Unit	Topics	Springboard Course	CO
1. Differential Calculus	Introduction to limits and continuity, Successive differentiation, Leibnitz's Theorem for n^{th} derivatives of the product of two functions, Rolle's theorem, Lagrange's Mean Value Theorem, Cauchy's Mean Value Theorem, Taylor Series Expansion, Maclaurin Series Expansion	<u>Calculus: Getting Started with Derivatives</u>	1
2. Partial Derivatives	Multivariable functions, Introduction to Partial derivatives, Differentiation treating one variable as constant, Composite functions, Chain Rule, Total Derivatives, Euler's Theorem on Homogeneous Functions, Applications, Jacobians, functional dependence/independence	<u>ML Algorithms: Multivariate Calculation & Algorithms</u>	2